

University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2024
Course Title: Data Structures and Algorithms I
Course Code: CSE 211

Marks: 20

Time: 1(one) hour

(Answer all questions)

Q.No.		Marks
1a	Briefly explain the classification of non-primitive data structure with an example.	3
1b	Suppose you are designing a memory layout for a 2D array in row-major and column major order. The base address of the array is 3000, and the array dimensions are 4x3, with each element being of type float (4 bytes). Calculate the memory location of the element at location (3, 2) for both types of array representation.	3
1c	Interpret the meaning of Big-O, Big- Ω (Omega), and Big- Θ (Theta) notations. How does Big-O notation describe the worst-case scenario for an algorithm?	4
2a	Describe the memory representation of Linked List.	2
2b	Write an algorithm to find out the summation of diagonal elements of a two dimensional array.	3
2c	Figure 1 shows a linked list of integer type values.	5



Figure 1: Linked List

Write down algorithms for the following tasks based on this sequence:

- To insert a node of 8 after the node of 7.
- To delete the node with value 2.